



Session friendly spawning with MPI

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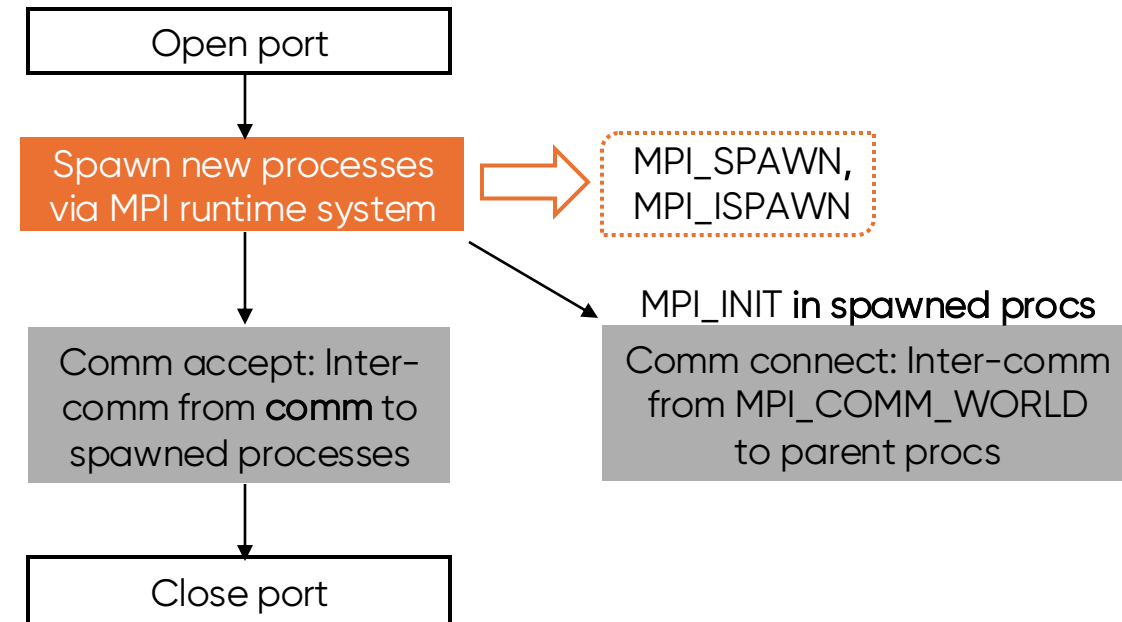
Motivation

- Lack of support for Session friendly spawning
 - Since MPI 4.0
 - At least spawned MPI processes must use MPI World model for synchronization with parent and creation of inter-communicator in MPI_INIT
- Lack of non-blocking spawning with MPI
- Increasing interest in malleability/ elasticity with MPI
 - MPI Sessions are good foundation
 - No dependency on static world comm by definition

Proposal (1)

- Idea
 - Spawn new processes independent of MPI comm; then use MPI procedures to create inter-comm
 - Resolve need for sync. via MPI_SESSION_INIT (local operation!), defer to connection establishment
- MPI_SPAWN and MPI_ISPAWN
 - Spawn new processes, no connection establishment
 - Blocking and non-blocking variant
- MPI_SPAWN_TEST_PARENT
 - Counterpart to MPI_COMM_GET_PARENT
 - Find out if an MPI process has a parent MPI process
- Local operations, no MPI comm involved!

MPI_COMM_SPAWN(_MULTIPLE)
collective over input comm



Proposal (2)

- Idea
 - Spawn new processes independent of MPI comm; then use MPI procedures to create inter-comm
 - Resolve need for sync. via MPI_SESSION_INIT (local operation!), defer to connection establishment
- MPI_SPAWN and MPI_ISPAWN
 - Spawn new processes, no connection establishment
 - Blocking and non-blocking variant
- MPI_SPAWN_TEST_PARENT
 - Counterpart to MPI_COMM_GET_PARENT
 - Find out if an MPI process has a parent MPI process
- Local operations, no MPI comm involved!

```
MPI_SPAWN(int count, char **array_of_commands,  
          char ***array_of_argv,  
          const int *array_of_maxprocs,  
          MPI_Info *array_of_info)
```

```
MPI_ISPAWN(int count, char **array_of_commands,  
           char ***array_of_argv,  
           const int *array_of_maxprocs,  
           MPI_Info *array_of_info,  
           MPI_Request *request)
```

```
MPI_SPAWN_TEST_PARENT(int *flag)
```

Remarks

- Co-existence of new spawn procedures with existing ones
- New spawn procedures are compatible to MPI Session model and MPI World model
- Implementation in ParaStation MPI (MPICH-based) with usage examples
 - https://github.com/ParaStation/psmpi/blob/master/mpich2/test/mpi/parastation/spawn/spawn_ex.c
- Déjà vu from ~30 years ago?
 - MPI-2 Journal of development from 1997, Section 2, "Spawning Independent Processes"
 - <https://www.mpi-forum.org/docs/mpi-jd/mpi-20-jod.ps>
 - Some similarities but also differences to this proposal

Draft PR

- <https://github.com/mpiwg-sessions/mpi-standard/pull/54>



Thank You!

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